

HPDE

**CAREER PREVIEW PROGRAM
PROGRESS STATUS**

Project Name : Multi Cloud Based Performance Evaluation
of Deployed Clusters

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HPE



Project Title

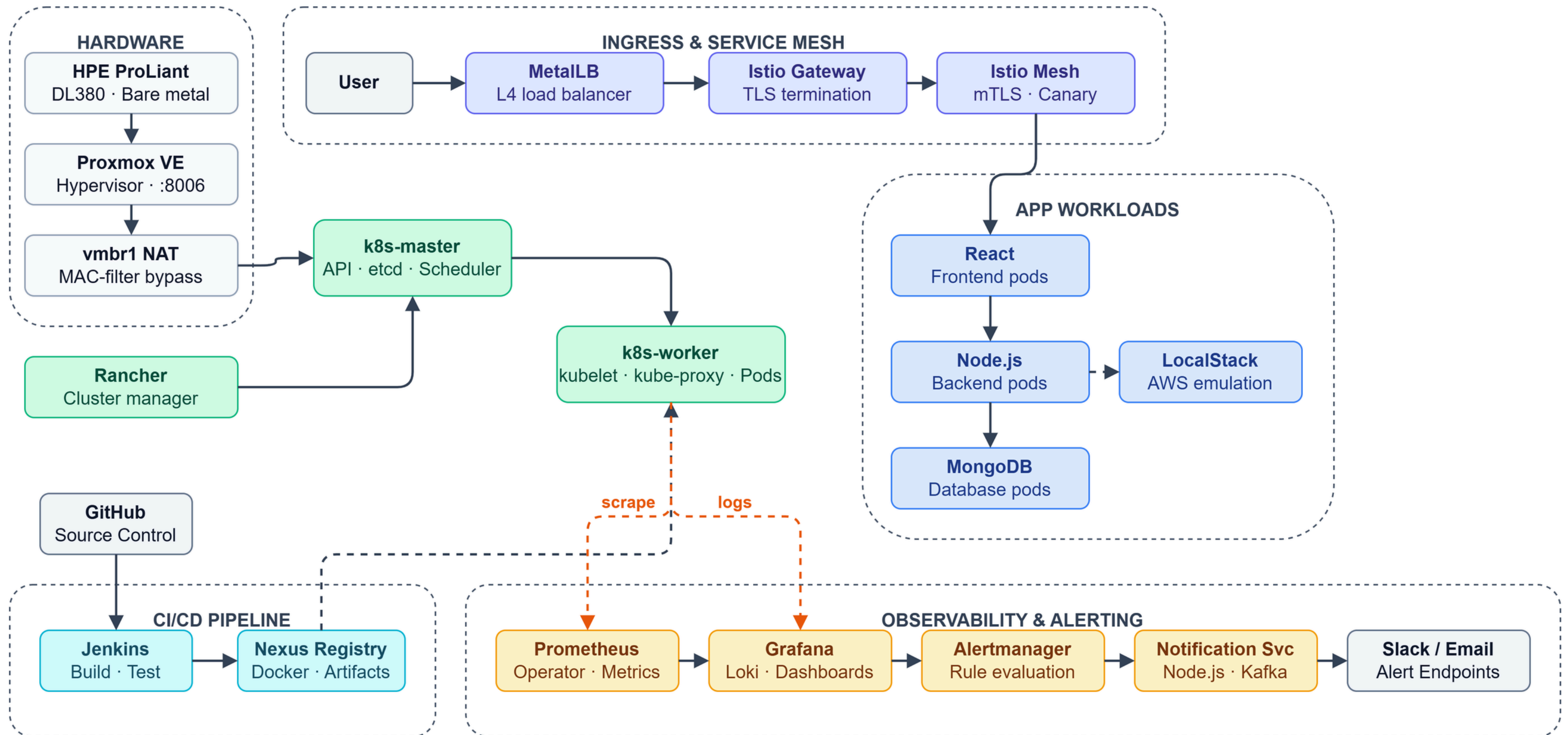


AIM - Multi Cloud Based Performance Evaluation of Deployed Clusters.

CORE OBJECTIVES:

- DevOps and Infrastructure Benchmarking in a Multi-Node Rancher Environment.
- Performance Evaluation and Resilience of Multi-Cloud Clusters on HPE ProLiant Architecture.
- Architecting and Benchmarking Hybrid Cloud Kubernetes Deployments.

High Level System Design



Infrastructure And Orchestration



Foundational Cluster Setup (Kubeadm):

- Bootstrapped an initial multi-node Kubernetes environment on standard lab hardware using a Type-2 hypervisor to rapidly prototype the application architecture.



Bare-Metal Load Balancing (MetalLB):

- Deployed MetalLB to dynamically assign external IP addresses to our services, effectively simulating cloud-provider load balancing capabilities within a local, bare-metal network.



Advanced Service Mesh Integration (Istio):

- Secured and observed pod-to-pod communications by injecting the Istio Service Mesh across our application workloads.
- Istio Ingress Gateway: Single entry point for edge traffic and dynamic routing.
- Canary Deployments: Leveraged Istio's weighted traffic routing for progressive delivery, safely shifting live user traffic between versions to validate stability before full cutover.



Infrastructure And Orchestration

HPE

Bare-Metal Virtualization (Proxmox VE):

- Migrated to a Type-1 hypervisor running headless Ubuntu Server VMs.
- Leveraged hardware CPU pass-through and VirtIO networking to ensure unpolluted metrics for multi-cluster evaluations.



Enterprise Cluster Management (Rancher & RKE2):

- Replaced manual kubeadm with a centralized, production-grade Rancher control plane managing our Master and Worker nodes.



Custom NAT Gateway Architecture:

- Security Bypass: Overcame strict campus Layer-2 MAC address filtering by using an internal virtual switch (vbr1).
- Network Isolation: Isolated the Kubernetes nodes on a secure, private subnet, routing all outbound internet traffic cleanly through a single whitelisted host IP.

Core Platform Services

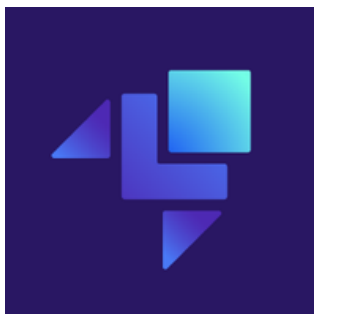


Notification Microservice (Custom Application):

- Architecture: Event-driven, stateless Node.js microservice utilizing Apache Kafka for reliable message brokering.
- Functionality: Employs a fan-out Pub/Sub model to instantly process incoming alert webhooks and broadcast them to enterprise endpoints (Slack, Email).

Cloud Emulation Service (LocalStack)

- Deployed within the Rancher cluster to emulate cloud-native services locally.
- Allows the team to test multi-cloud deployment scenarios and API integrations on the HPE ProLiant hardware without incurring external cloud costs.



Artifact & Registry Management (Sonatype Nexus)

- Acts as the centralized repository for the cluster.
- Securely hosts and manages our private Docker container images, npm packages, and build artifacts to ensure fast, reliable deployments across nodes.



Infrastructure Monitoring Service



Core Tooling: Prometheus (Metrics), Loki (Logs), Grafana (Dashboards), and Alertmanager.

Automated Setup: Deployed via Prometheus Operator, utilizing Custom Resource Definitions (CRDs) for dynamic target discovery across the Rancher/ProLiant cluster.

Comprehensive Alert Triggers:

- Infrastructure Health: High CPU/Memory pressure, pod crash loops, or node unavailability.
- Log Monitoring (Loki): Real-time pattern matching for critical error codes and exception stack traces within container logs.
- SLA Breaches: Alerts triggered by API latency spikes, dropped packets, or error rates exceeding strict Service Level Agreements.

Automated Notification Pipeline:

- Alertmanager routes triggered events via Webhook to our custom Notification Microservice (Node.js/Kafka).
- The fan-out architecture instantly broadcasts the alerts to designated Slack channels and Email endpoints.



Grafana



Quality Assurance & Testing (CI)

End-to-End Deployment Flow: Jenkins orchestrates the entire lifecycle by pulling approved code from GitHub, building Docker images, pushing them securely to our private Sonatype Nexus registry, and executing Helm chart upgrades to perform rolling pod updates across the Rancher cluster.

Automated Quality Gates:

- Early Validation: GitHub Actions runs ESLint on pull requests to enforce syntax consistency prior to merging.
- Security & Debt: SonarQube enforces a strict quality gate, halting the Jenkins build if vulnerabilities, bugs, or excessive code smells are detected.

Unit Testing: Vitest automatically validates the core business logic of our Node.js microservices, ensuring application reliability before containerization.

sonarqube



Security Integration - Admin

Centralized Identity Provisioning : Configured Keycloak as the Primary Identity Provider (IdP) to manage users, realms, and OIDC client protocols.

- Jenkins Programmatic Authentication - Implemented Jenkins authentication via JSON Service Accounts (Service Account Tokens) to enable secure, non-interactive API access for automated pipelines.
- Nexus Gateway Deployment - Deployed OAuth2-Proxy as a Kubernetes-native gateway to intercept all incoming traffic on Port 30004 and enforce OIDC validation.



The OIDC Handshake : User requests are redirected from the Proxy to Keycloak for credential verification, generating a secure JWT (JSON Web Token).



Nexus "Rut Auth" Validation : Configured the Nexus Remote User Token (Rut Auth) Realm to trust the Proxy's headers and bypass internal login prompts.

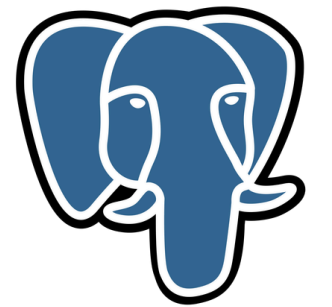
RBAC Synchronization : Mapped Keycloak identities to Nexus External Roles, granting administrative privileges (nx-admin) automatically based on the Proxy's identity.



Security Integration - Consumers

HPE

Secure Registration & Onboarding : Enabled controlled signup flows with Just-In-Time (JIT) synchronization of authenticated users into MongoDB.



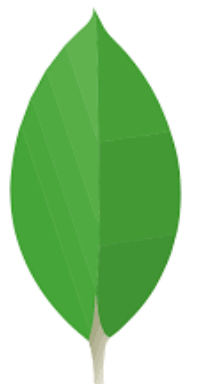
OIDC Federation & SSO : Integrated OpenID Connect (OIDC) between React frontend and Node.js backend for secure Single Sign-On and token sessions.

JWT Access Enforcement : Validated Bearer Tokens at API gateways to ensure only authenticated users access protected services.



Role-Based Access Control (RBAC) :

- Implemented student, alumni, mentor, recruiter, and admin roles with least-privilege authorization.
- Governed Approval Workflow - Added admin approval lifecycle where new users remain pending until approved or rejected.



Persistent Identity Mapping : Used immutable Keycloak Subject ID (UUID) for reliable identity correlation across systems.

API Documentation

Dedicated Documentation Route

A standalone /docs route is served as part of the application providing a single entry point for all API references, onboarding guides, and developer resources. This ensures contributors and reviewers can navigate the platform's contracts without digging into source code.

Swagger UI + OpenAPI Specifications

All microservice endpoints are documented via OpenAPI 3.x specifications and rendered interactively through Swagger UI - allowing the team to inspect request/response schemas, test endpoints live, and validate API contracts without external tooling.

MkDocs Structured Project Documentation

Leveraged MkDocs to build a route-driven documentation portal where each section of the application is represented through dedicated pages mapped to relevant modules and workflows. Maintained content through structured Markdown files, enabling scalable updates for architecture details, feature flows, setup guides, ownership mapping, deployment runbooks, and operational documentation.

Project Plan & Roadmap

Load Simulation : Executing automated stress tests against the deployed microservices to simulate high-traffic enterprise environments.

Network & Performance Analysis: Utilizing our observability stack to capture and evaluate critical multi-cloud metrics, focusing specifically on packet loss, jitter, latency, and throughput between on-premise and cloud nodes.

Dynamic Auto-Scaling Benchmarks: Configure Horizontal Pod Autoscalers (HPA) for your core services. During your load simulation, track exactly how long it takes for the cluster to provision new pod replicas across the multi-cloud boundary when traffic spikes, and evaluate if the Prometheus metrics reflect the scaling accurately.

Hybrid Cloud Expansion: Extending the Rancher cluster to incorporate external public cloud nodes (e.g., Oracle Cloud Infrastructure or AWS) to finalize the multi-cloud topology.



**THANK
YOU**